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```
# generate validation data
valid_generator = datagen.flow_from_dataframe(
    dataframe=train_data,
    directory=train_path,
    x_col="id",
    y_col="label",
    subset="validation",
    batch_size=BATCH_SIZE,
    seed=RANDOM_STATE,
    class_mode="binary",
    target_size=(64, 64) # original image = (96, 96)
```

```
Found 33003 validated image filenames belonging to 2 classes.
```

```
# Setup GPU accelerator - configure Strategy. Assume TPU...if not set default for GPU/CPU
tpu = None
try:
    tpu = tf.distribute.cluster_resolver.TPUClusterResolver()
    tf.config.experimental_connect_to_cluster(tpu)
    tf.tpu.experimental.initialize_tpu_system(tpu)
    strategy = tf.distribute.TPUStrategy(tpu)
except ValueError:
    strategy = tf.distribute.get_strategy()
```

Start First model building and training

Here we build our first model. This model is fairly simple with a handful of layers, void of activations except the last layer. The last layer we use a sigmoid activation function since our data is binary (0-1). We can see below how the model is built and how many parameters we will need to train for this model. The next cell below we can see the model training per epoch.

Show hidden code

2022-11-10 15:59:21.738049: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero

2022-11-10 15:59:21.738763: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero

2022-11-10 15:59:21.741335: I tensorflow/core/platform/cpu_feature_guard.cc:142] This TensorFlow binary is optimized with oneAPI Deep Neural Network Library (oneDNN) to use the following CPU instructions in performance-critical operations: AVX2 AVX512F MA

To enable them in other operations, rebuild TensorFlow with the appropriate compiler flags.

Model: "sequential"		
Layer (type)	Output Shape	Param #
=====		
conv2d (Conv2D)	(256, 62, 62, 16)	448
conv2d_1 (Conv2D)	(256, 68, 68, 16)	2320
max_pooling2d (MaxPooling2D)	(256, 38, 38, 16)	0
conv2d_2 (Conv2D)	(256, 28, 28, 32)	4640
conv2d_3 (Conv2D)	(256, 26, 26, 32)	9248
average_pooling2d (AveragePo	(256, 13, 13, 32)	0
flatten (Flatten)	(256, 5408)	0
dense (Dense)	(256, 1)	5409
=====		
Total params: 22,865		
Trainable params: 22,865		
Non-trainable params: 0		

```

2022-11-10 15:59:21.741665: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.742593: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.743466: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.744340: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.745213: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.746087: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.746960: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.747834: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.748707: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.749581: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.750454: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.751328: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.752201: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2022-11-10 15:59:21.753075: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1518] Created device /job:localhost/repllica/0/task:0/device:GPU:0 with 15381 MB memory :> device 0, name: Tesla P100-Pcie-16GB, pci bus id: 8000:08:04:0c, compute capability: 6.0

```

```
EPOCHS = 10

# train the model
history_model_one = model_one.fit_generator(
    train_generator,
    epochs = EPOCHS,
    validation_data = valid_generator)
```

```
/opt/conda/lib/python3.7/site-packages/keras/engine/training.py:1972: UserWarning:
```

`Model.fit_generator` is deprecated and will be removed in a future version. Please use `Model.fit`, which supports generators.

```
2022-11-10 15:59:26.782941: I tensorflow/compiler/mlir/mlir_graph_optimization_pass.cc:185] None
of the MLIR Optimization Passes are enabled (registered 2)
```

Epoch 1/18

```
2022-11-10 15:59:30.841650: I tensorflow/stream_executor/cuda/cuda_dnn.cc:369] Loaded cuDNN versi
on 8005
```

```
731/731 [=====] - 1696s 2s/step - loss: 0.6029 - accuracy: 0.6867 - auc: 0.7268 - val_loss: 0.5320 - val_accuracy: 0.7523 - val_auc: 0.8085
```

```
731/731 [=====] - 292s 399ms/step - loss: 0.5450 - accuracy: 0.7398 - au  
c: 0.7933 - val loss: 0.5365 - val accuracy: 0.7534 - val auc: 0.8256
```

Epoch 3/10
731/731 [=====] - 289s 395ms/step - loss: 0.5174 - accuracy: 0.7565 - au

Epoch 4/10
731/731 [=====] - 284s 388ms/step - loss: 0.4950 - accuracy: 0.7697 - au

```
Epoch 5/10
731/731 [-----] - 283s 388ms/step - loss: 0.4795 - accuracy: 0.7701 - au
```

```
c: 0.8451 - val_loss: 0.4807 - val_accuracy: 0.7794 - val_auc: 0.8567
Epoch 6/10
```

```
c: 0.8516 - val_loss: 0.4606 - val_accuracy: 0.7903 - val_auc: 0.8688
Epoch 7/10
```

```
7317/731] [=====] - 290s 39/ms/step - loss: 0.4636 - accuracy: 0.7883 - au  
c: 0.8559 - val_loss: 0.4716 - val_accuracy: 0.7829 - val_auc: 0.8661  
Epoch 8/10
```

```
731/731 [=====] - 284s 388ms/step - loss: 0.4574 - accuracy: 0.7917 - au
c: 0.8602 - val_loss: 0.4388 - val_accuracy: 0.8014 - val_auc: 0.8727
```

```
731/731 [=====] - 289s 396ms/step - loss: 0.4526 - accuracy: 0.7945 - au
c: 0.8634 - val_loss: 0.4595 - val_accuracy: 0.7936 - val_auc: 0.8681
```

```
Epoch 10/10
731/731 [=====] - 278s 380ms/step - loss: 0.4490 - accuracy: 0.7966 - au
```

Now that we have trained the model, we can take a look at how it did graphically with the training data. Below we can see the accuracy and loss with regards to the validation and training set.

Show hidden code

